F FWallSim · Firewall Course

CLASS 11

# SSL-VPN for remote workers

SSL-VPN settings, portals, ssl.root policy

Advanced | ~25 minutes | FWallSim v1.0.0 (FortiOS 7.4 style)

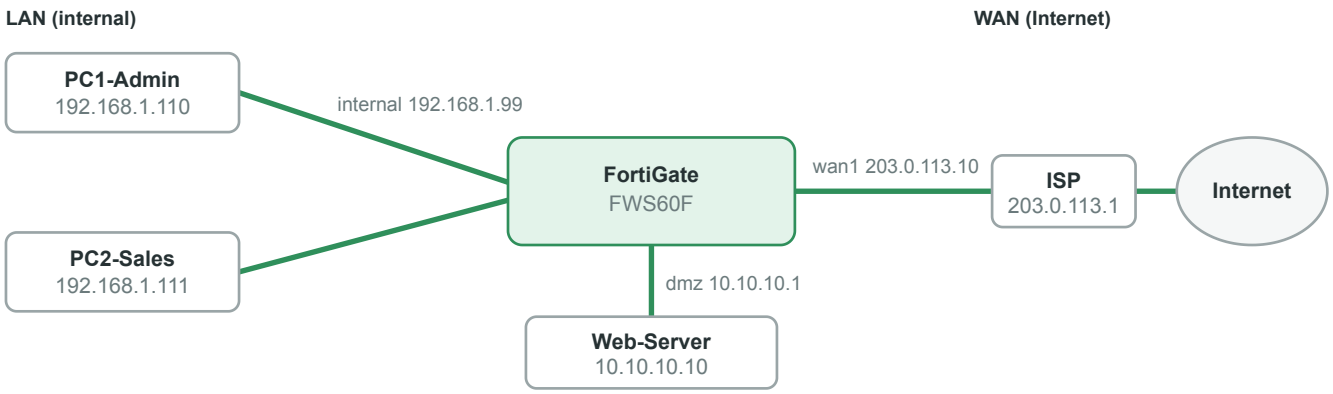
### YOU WILL LEARN

- SSL-VPN settings
- portals
- ssl.root policy

### BEFORE YOU START

Finish Class 10 first.

## The lab network




## The goal

### THE IDEA

Remote staff need FortiClient access to the office PCs. Create a local user, a group, SSL-VPN settings on wan1 and the ssl.root policy, then connect Remote-Laptop.

## How to start

1. Open FWallSim and log in.
2. Click **Labs** in the header.
3. Find **SSL-VPN for remote workers** and click **Start**.
4. The Lab Guide panel opens. Each task below gets a green tick as soon as it works.

## Tasks

1. **SSL-VPN listens on wan1 on a port other than the admin HTTPS port**

   VPN > SSL-VPN Settings: Listen on wan1, port 10443.

2. **A local user is in a user group**

   User & Authentication > User Definition > Create New (Local User, with a password). Then User Groups > Create New (Firewall) and add the user as a member.

3. **An authentication rule maps that group to a tunnel-mode portal**

   VPN > SSL-VPN Settings > Authentication/Portal Mapping > Create New: your group, portal tunnel-access. That portal uses split tunneling, so only office traffic goes through the VPN and the laptop reaches the internet directly.

4. **A policy ssl.root → internal allows that group**

   Firewall Policy > Create New: Incoming SSL-VPN tunnel interface (ssl.root), Outgoing internal, Source all + your group, Destination the office LAN, Service ALL, ACCEPT.

5. **Remote-Laptop is connected with FortiClient**

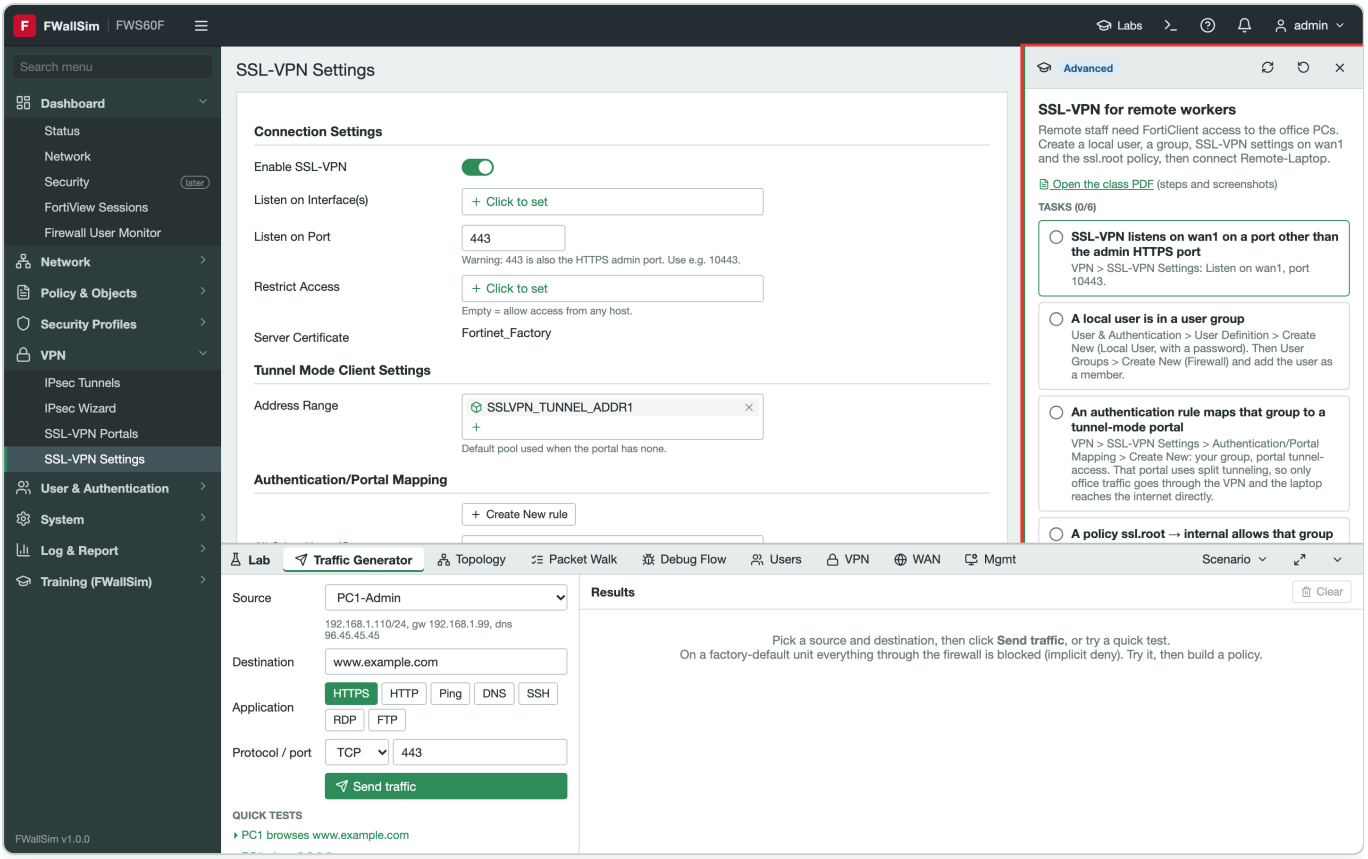
   Lab > VPN: connect Remote-Laptop to 203.0.113.10 on your SSL-VPN port with the local user.

6. **Remote-Laptop can RDP to PC1 through the tunnel**

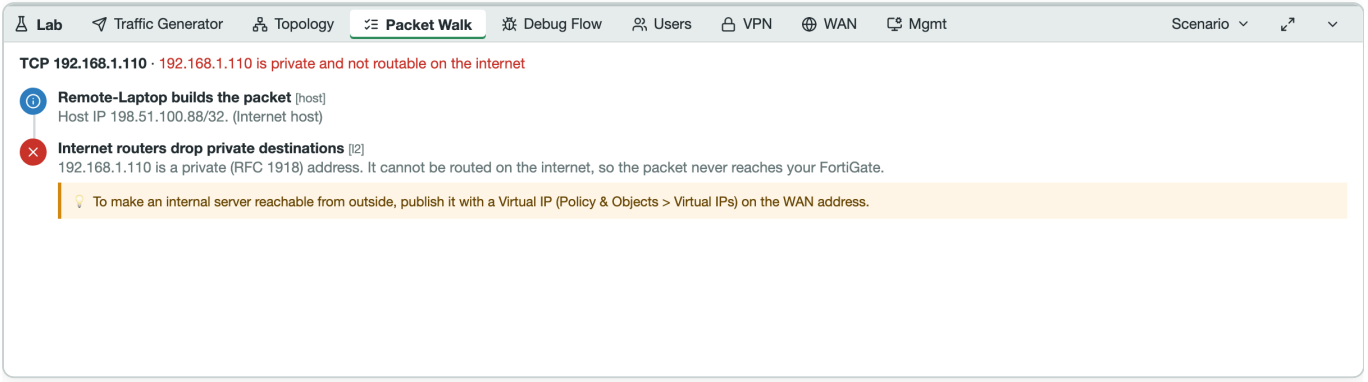
   Send Remote-Laptop → 192.168.1.110 TCP/3389.

## Screenshots

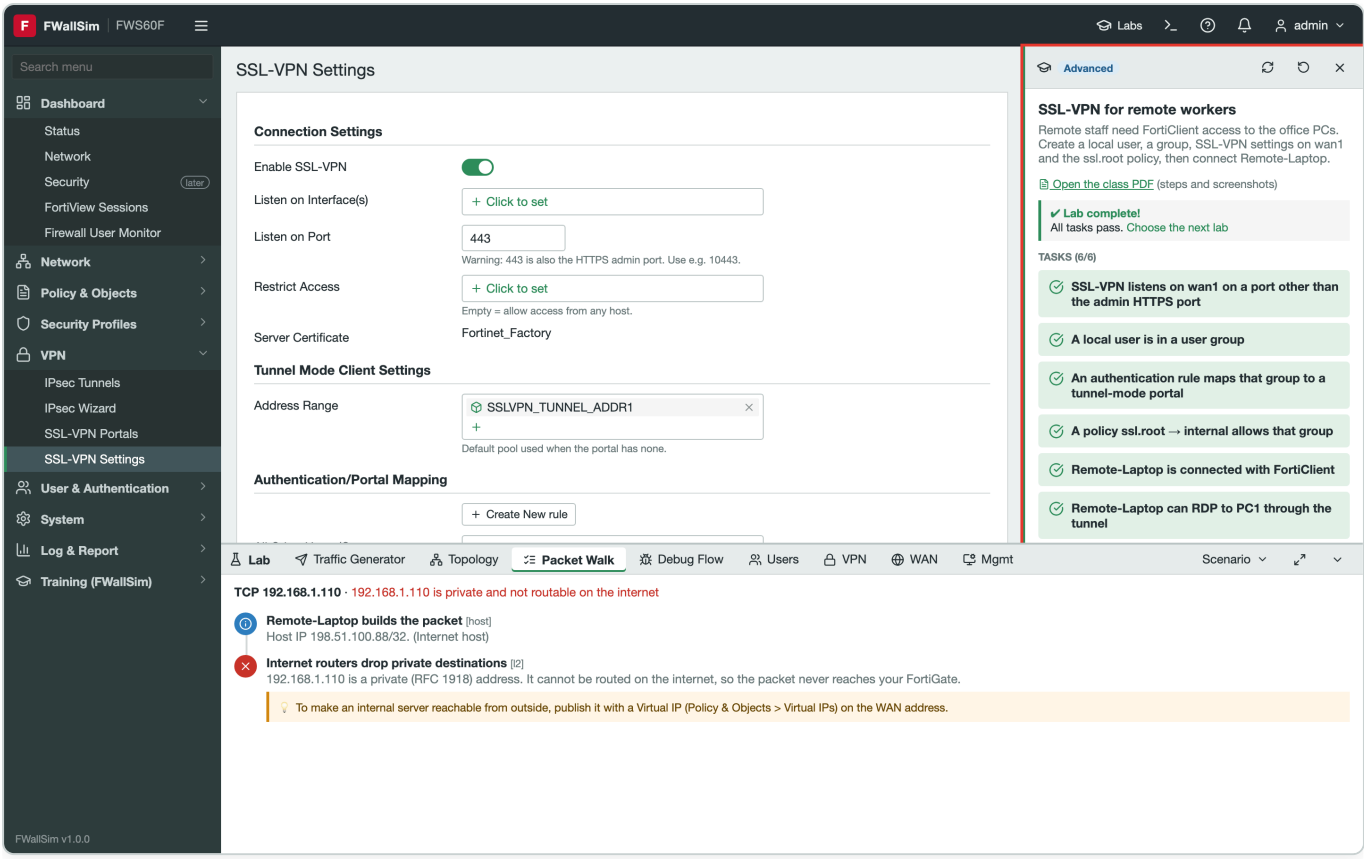
Numbered red markers show where to click or what to look at.



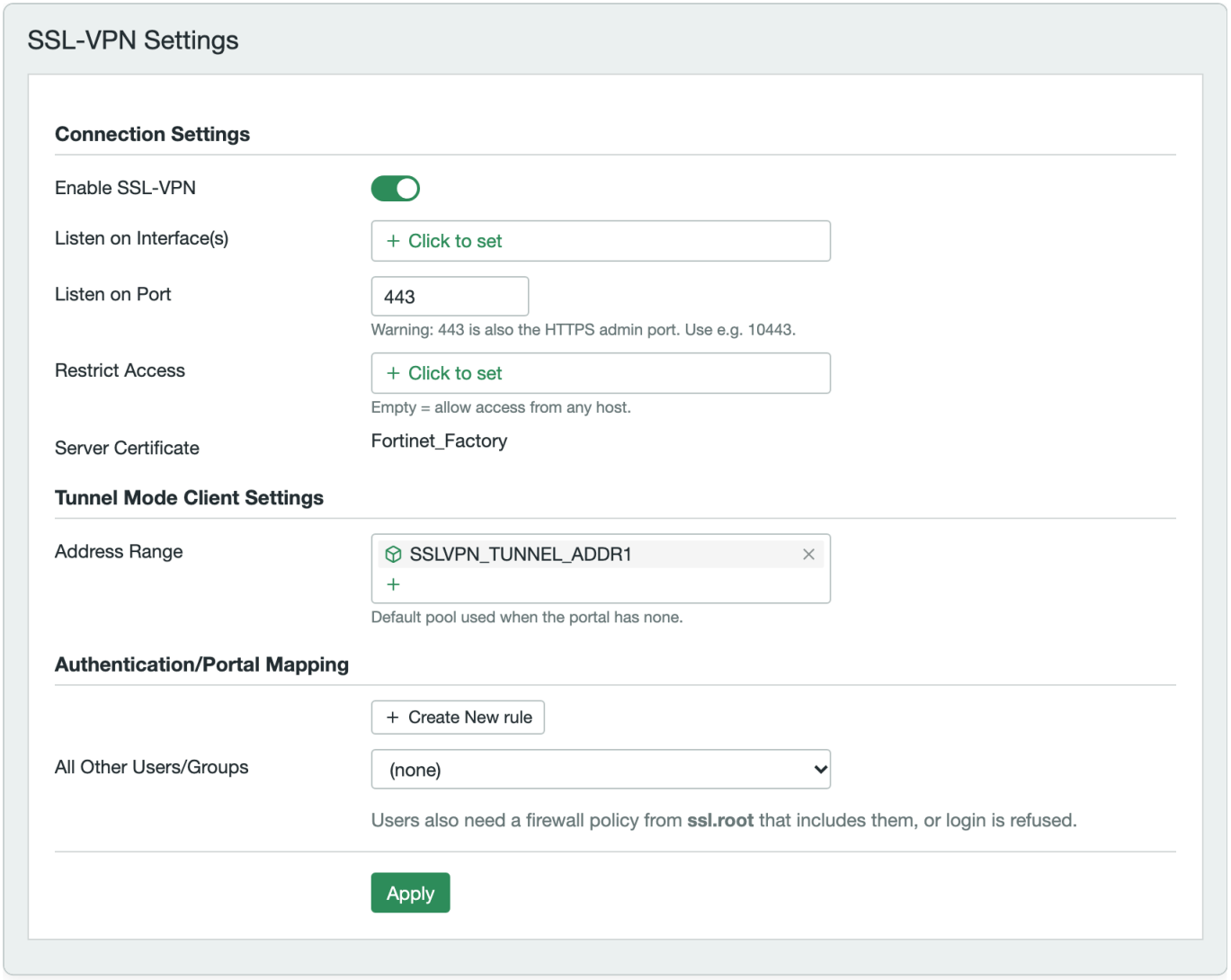

*Lab Started*



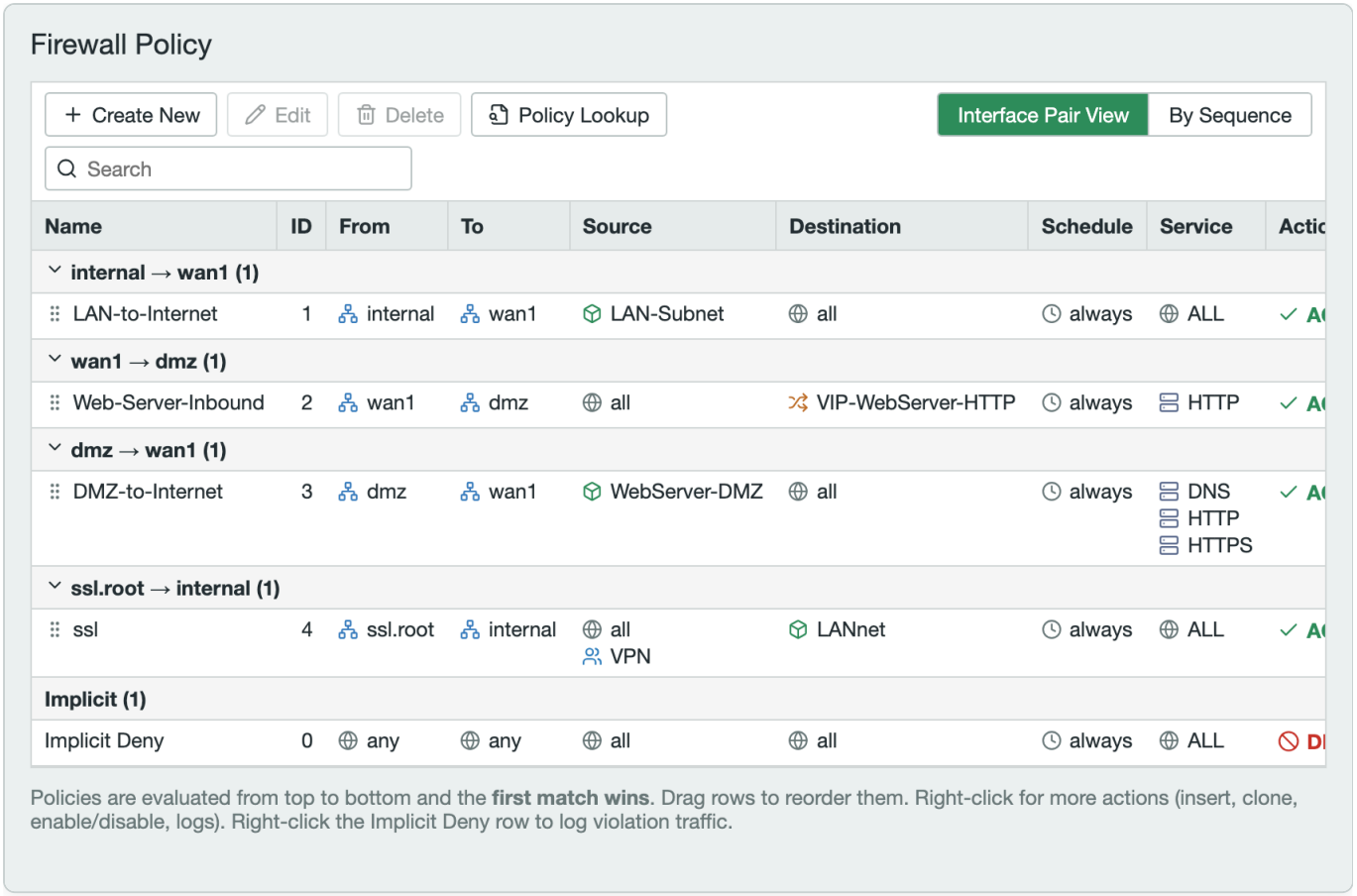

*Packet Walk Before*


FWallSim | FWS60F
Labs
admin
Search menu
Dashboard
Status
Network
Security
later
FortiView Sessions
Firewall User Monitor
Network
Policy & Objects
Security Profiles
VPN
IPsec Tunnels
IPsec Wizard
SSL-VPN Portals
SSL-VPN Settings
User & Authentication
System
Log & Report
Training (FWallSim)
FWallSim v1.0.0
SSL-VPN Settings
Connection Settings
Enable SSL-VPN
Listen on Interface(s)
+ Click to set
Listen on Port
443
Warning: 443 is also the HTTPS admin port. Use e.g. 10443.
Restrict Access
+ Click to set
Empty = allow access from any host.
Server Certificate
Fortinet_Factory
Tunnel Mode Client Settings
Address Range
SSLVPN_TUNNEL_ADDR1
Default pool used when the portal has none.
Authentication/Portal Mapping
+ Create New rule
Advanced
SSL-VPN for remote workers
Remote staff need FortiClient access to the office PCs. Create a local user, a group, SSL-VPN settings on wan1 and the ssl.root policy, then connect Remote-Laptop.
Open the class PDF (steps and screenshots)
✔ Lab complete!
All tasks pass. Choose the next lab
TASKS (6/6)
SSL-VPN listens on wan1 on a port other than the admin HTTPS port
A local user is in a user group
An authentication rule maps that group to a tunnel-mode portal
A policy ssl.root → internal allows that group
Remote-Laptop is connected with FortiClient
Remote-Laptop can RDP to PC1 through the tunnel
Lab
Traffic Generator
Topology
Packet Walk
Debug Flow
Users
VPN
WAN
Mgmt
Scenario
TCP 192.168.1.110 · 192.168.1.110 is private and not routable on the internet
Remote-Laptop builds the packet [host]
Host IP 198.51.100.88/32. (Internet host)
Internet routers drop private destinations [l2]
192.168.1.110 is a private (RFC 1918) address. It cannot be routed on the internet, so the packet never reaches your FortiGate.
To make an internal server reachable from outside, publish it with a Virtual IP (Policy & Objects > Virtual IPs) on the WAN address.

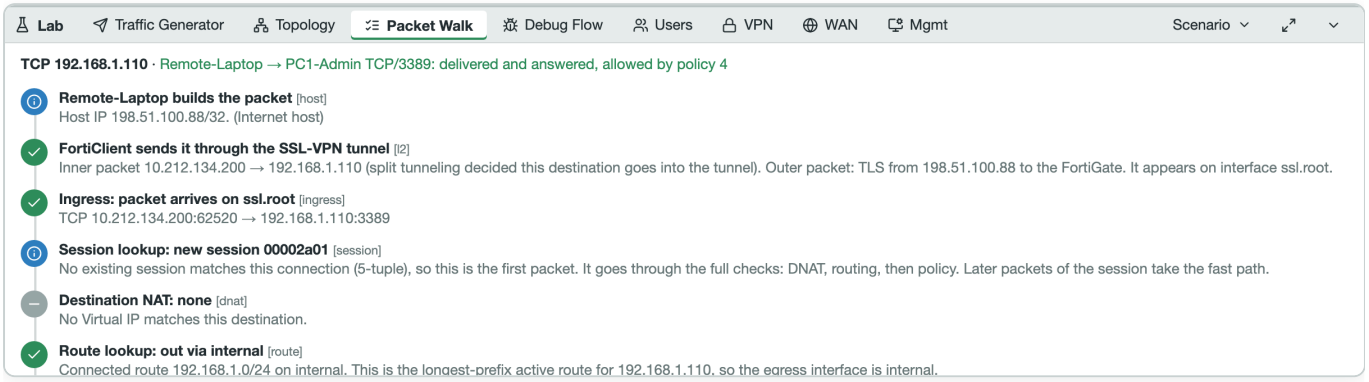
*Lab Solved*




*SSL-VPN Settings*




*Firewall Policy*




*Packet Walk After*

**Next:** Class 12: SD-WAN with two ISPs